

1 Project Title

From VAD to Learned Speech-Presence Probability for Adaptive Speech Enhancement

2 Student Information

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3 Project Overview

This project aims to address the fundamental limitation of classical speech enhancement systems in highly dynamic, non-stationary acoustic environments. Specifically, it investigates how speech-presence should be accurately estimated so that the noise Power Spectral Density (PSD) can continue to adapt when speech and noise overlap, and the noise characteristics change abruptly. The proposed solution involves a highly controlled comparative study of three distinct speech-presence estimation paradigms: (1) Hard frame-level Voice Activity Detection (VAD), (2) Classical statistical soft Speech-Presence Probability (SPP) via Improved Minima Controlled Recursive Averaging (IMCRA), and (3) Modern Pretrained Temporal Convolutional Network (TCN) Learned SPP. All three estimators will drive a fixed Wiener filter backend to ensure deep scientific interpretability, isolating the specific impact of the noise-tracking methodology on overall enhancement performance.

4 Background and Motivation

In conventional speech enhancement algorithms, a standard frame-based VAD makes binary decisions (speech or noise). Consequently, the noise estimate is typically updated only during frames labeled as purely noise. If a speaker continues talking for several seconds while the background noise changes underneath the speech—such as an abrupt shift from fan noise to street traffic, or a sudden change in Signal-to-Noise Ratio (SNR)—the noise estimator is left with no "safe" frame to update its parameters. This structural limitation causes the enhancement algorithm to apply outdated noise profiles, severely degrading speech quality. Modern research, including recent 2026 studies on learning-based *a posteriori* SPP estimation, indicates that estimating a probabilistic SPP at every time-frequency bin enables continuous, soft noise PSD updates. However, completely replacing a digital signal processing (DSP) pipeline with an end-to-end neural network obscures the underlying mechanisms. Therefore, this project is highly motivated to construct a transparent experimental framework where the enhancement formula remains constant, but the driving statistical or learned intelligence varies, thereby cleanly answering whether modern robust performance comes from accurate speech state estimation or raw neural modeling.

5 Proposed Methodology

The project will be strictly developed and simulated within the MATLAB environment, bridging fundamental DSP techniques with modern inference baselines.

- **Signal Processing Framework:** The core architecture relies on Short-Time Fourier Transform (STFT), *a priori* and *a posteriori* SNR estimation, recursive noise PSD updating, Wiener gain application, and Inverse STFT with overlap-add reconstruction.
- **System Implementations:**
 - *System A:* Energy-based Hard VAD baseline (updating noise only during total silence).
 - *System B:* IMCRA baseline utilizing classical probabilistic speech-

presence rules to continuously update likely-noise bins.

- *System C*: A pretrained TCN SPP estimator acting as a modern reference to track time-frequency SPP using learned local/global spectral structures.
- *Diagnostic Oracle System*: Using the ground-truth separated noise to construct an absolute upper bound for PSD estimation.
- **Dataset Formulation**: Clean speech will be sourced from LibriSpeech or TIMIT. Noise sources will be drawn from standard environmental datasets (e.g., DEMAND, MUSAN). Instead of standard stationary mixtures, the methodology relies heavily on custom-designed *dynamic transitions*—specifically crafting mixtures where noise levels (e.g., 10 dB dropping to 0 dB) or noise types (e.g., fan to traffic) change precisely at known timestamps (e.g., $t = 3$ seconds) to rigorously test the adaptation speed of the estimators.

6 Expected Outcomes

The project is expected to deliver a fully functional, highly documented MATLAB simulation suite. Quantitatively, the project will output detailed comparative metrics that move beyond standard global SNR improvements. The expected performance metrics include:

- **Noise PSD Estimation Error**: Direct calculation of the logarithmic spectral error between the estimated noise PSD and the true Oracle noise PSD, representing the core efficacy of the estimators.
- **Adaptation Time (Settling Time)**: The exact temporal duration (measured in milliseconds) required for the system to recover and push the PSD error below a predetermined threshold following an abrupt environmental shift.
- **SI-SDR Improvement & STOI**: To evaluate the final perceptual and intelligibility outcomes of the enhanced speech signals.

- **Computational Complexity:** A practical analysis of the real-time factor and processing latency required by classical statistical methods versus neural learned methodologies.

All code, configuration metadata, and representative audio samples will be comprehensively hosted on the GitHub repository.

7 Timeline (Weeks 1-13)

- **Week 1-2:** Consider Project Topic and redefine the core research question toward continuous PSD tracking.
- **Week 3-5:** Conduct literature review on classical IMCRA and modern Learned SPP architectures. Synthesize controlled dynamic noise datasets with precise transition timestamps.
- **Week 6-9:** Initial implementation and testing. Develop the STFT/ISTFT pipeline and the System A (Hard-VAD Wiener) baseline. Validate failure modes in continuous speech scenarios.
- **Week 10-11:** Integrate System B (IMCRA) and System C (Pretrained TCN SPP inference wrapper). Implement Oracle diagnostic bounds. Execute optimization and batch evaluation across all metrics (PSD error, adaptation time, SI-SDR, STOI).
- **Week 12-13:** Compile final comparative data, generate time-frequency SPP heatmaps and adaptation curves. Finalize the academic report and perfect the GitHub repository documentation.

8 References

1. Tao, Y. et al. (2026). Learning-based a posteriori speech presence probability estimation and applications. *Journal on Audio, Speech, and Music Processing*.
2. Cohen, I. (2003). Noise spectrum estimation in adverse environments:

Improved minima controlled recursive averaging. *IEEE Transactions on Speech and Audio Processing*, 11(5), 466-475.

3. Ephraim, Y., & Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 32(6), 1109-1121.

9 AI Usage Declaration

Google Gemini was used as a supplementary tool during the preparation of this project proposal and preliminary MATLAB coding. Its use was limited to brainstorming the specific research direction, suggesting a proposal structure aligned with course guidelines, assisting with MATLAB syntax for dynamic noise generation, and providing language-editing suggestions for the proposal text.

Prompts included:

1. “What would be a more modern and research-focused direction to upgrade a classical VAD-based Wiener filter system in non-stationary noise environments?”
2. “Write a MATLAB script to synthesize a dynamic noise mixture with an abrupt SNR level transition (e.g., 10dB to 0dB) at a specific timestamp.”
3. “Review this project overview paragraph and suggest improvements to its grammar, clarity, and academic expression without changing its technical meaning.”

The student independently defined the final research scope, conducted the literature review on speech-presence probability (SPP) and IMCRA, executed and verified all MATLAB simulations, and analyzed the acoustic failure cases. Any AI-generated code snippets or text suggestions were critically reviewed, debugged line-by-line, and substantially adapted to meet the specific requirements of the project. AI-generated material was not treated as an authoritative or citable academic source. The student

takes full responsibility for the accuracy, code execution, referencing, and final content of this submission.